

D679

Elementary Science and Engineering Methods

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A third-grade teacher who was a park ranger before teaching is starting a unit on weather and climate. The district just bought new science notebooks. Which launch best fits three-dimensional, phenomenon-driven teaching?

- A. Have students copy weather vocabulary into the new notebooks.
- B. Read the textbook chapter aloud, then show the time-lapse at the end.
- C. Give a quiz on weather facts.
- D. Show a time-lapse of a puddle on the playground that is gone by afternoon and ask students what they notice and wonder.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A fourth-grade teacher in a school that recently won a STEM award is planning a 5E lesson on circuits. She has already engaged students with a flashlight that will not light. What should come next?

- A. The teacher explains complete circuits with a diagram.
- B. Students apply circuits to design a model house with lights.
- C. Students take a unit test on circuits.
- D. Students explore with batteries, wires, and bulbs to find ways to make the bulb light.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

In a 5E lesson, students use what they learned about insulation to design a lunch box that keeps ice from melting. Which phase is this?

- A. Elaborate.
- B. Evaluate.
- C. Engage.
- D. Explore.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A second-grade class with a new maker cart is asked to build a bridge from craft sticks that must span 30 cm and hold a toy car. Students may use only 20 sticks and one bottle of glue. The class earned a pizza party last week. Which statement correctly describes the task?

- A. It is engineering, and it has no constraints because the maker cart is well stocked.
- B. It is an engineering task: spanning 30 cm and holding the car are criteria, and the limited sticks and glue are constraints.
- C. The 20-stick limit is a criterion for success.
- D. It is a science investigation because students use materials.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A teacher wants a science question rather than an engineering problem. Which fits?

- A. Why do some objects sink while others float?
- B. How can we build a boat that carries 50 pennies?
- C. Which recycled material makes the best bird feeder?
- D. How can we keep the classroom plants watered over break?

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Fifth graders at a school with a large garden want to find whether fertilizer helps bean plants grow taller. One group gives Plant A fertilizer and puts it by the window; Plant B gets no fertilizer and sits in a closet. The group's poster won a ribbon last year. What is the main problem with the plan?

- A. There is no problem; the plants are treated differently as planned.
- B. They need more plants, so one plant per group is always useless.
- C. They should use plants from the garden instead of pots.
- D. Two variables change, so any height difference cannot be traced to the fertilizer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Students test whether ramp height affects how far a toy car rolls. Which should they keep the same?

- A.** Nothing; real scientists change everything at once.
- B.** The distance the car rolls.
- C.** The car, the floor surface, and how the car is released.
- D.** Ramp height.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

During a pond study on a warm spring morning, a first grader writes several notes. The class is using new hand lenses. Which note is an observation rather than an inference?

- A.** The water is green because it is dirty.
- B.** The water is green, and I see tiny things moving in it.
- C.** The frogs must be hiding because they are scared of us.
- D.** The hand lenses make things bigger because they are magic.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A student observes that the sidewalk is wet and the sky is gray. Which statement is an inference based on these observations?

- A.** It probably rained recently.
- B.** It will certainly rain all week.
- C.** The sky is gray.
- D.** The sidewalk is wet.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

During a discussion about why ice melts faster on a metal tray than a plastic one, a third grader says, "The metal is colder, so it melts ice faster." The teacher, who has taught for 15 years, has a bell to signal cleanup. Which response best supports productive talk?

- A.** Ring the bell and start cleanup.
- B.** "Good job!" then call on the next student.
- C.** "What makes you think that? Does anyone want to add on or see it differently?"
- D.** "No, that's wrong. Metal conducts heat."

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A fifth-grade class in a new building with modern lab tables will launch small rubber-band rockets made of paper. The teacher has done similar activities for years and no one has been hurt. What is the best safety step?

- A.** Skip goggles since no one has been hurt before.
- B.** Require safety goggles for everyone, set a launch zone, and review rules before any launches.
- C.** Add goggles only if someone gets hit.
- D.** Cancel all hands-on science, because it is always unsafe.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Before a unit on seasons, a fourth-grade teacher who loves astronomy asks students to explain why summer is warmer. Many say Earth is closer to the sun in summer. The class has 24 students. What should the teacher do next?

- A.** Tell students they are wrong and state the correct answer.
- B.** Assign a reading on astronomy careers.
- C.** Record students' ideas, then use models and data (such as seasons being opposite in each hemisphere) to test them.
- D.** Ignore the ideas, since they will fade on their own.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A third-grade teacher ~~who was a park ranger before teaching~~ is starting a unit on weather and climate. ~~The district just bought new science notebooks.~~ Which launch best fits three-dimensional, phenomenon-driven teaching?

Target word: unit launch **Decides:** Anchor with a puzzling phenomenon students work to explain.

A	Camouflage	Uses the notebooks; vocabulary-first is not phenomenon-driven.
B	Inverted operation	Places the phenomenon last, as a demonstration of the answer.
C	Dropped element	Drops the phenomenon and student sensemaking.
D	Correct	A real phenomenon drives student questions and investigations.

2. Answer D Concept 2

A fourth-grade teacher ~~in a school that recently won a STEM award~~ is planning a 5E lesson on circuits. She has already engaged students with a flashlight that will not light. What should come next?

Target word: next phase after Engage **Decides:** Explore comes before Explain.

A	Inverted operation	Explain before Explore reverses the 5E model.
B	Confusion set	That is Elaborate, which follows Explain.
C	Adjacent category	Evaluate comes last.
D	Correct	Explore gives the experience the explanation will build on.

3. Answer A Concept 2 · transfer

In a 5E lesson, students use what they learned about insulation to design a lunch box that keeps ice from melting. Which phase is this?

Target word: name the phase from the activity **Decides:** Applying a concept to a new situation.

A	Correct	Students extend and apply the concept in a new context.
B	Adjacent category	Evaluation checks understanding; this task applies it.
C	Confusion set	Engage sparks interest at the start.
D	Inverted operation	Explore happens before the concept is explained.

4. Answer B Concept 3

A second-grade class with a new maker cart is asked to build a bridge from craft sticks that must span 30 cm and hold a toy car. Students may use only 20 sticks and one bottle of glue. ~~The class earned a pizza party last week.~~ Which statement correctly describes the task?

Target word: science or engineering, and its parts **Decides:** Designing a solution with criteria and constraints is engineering.

A	Camouflage	Keys on the maker cart; the stick and glue limits are constraints.
B	Correct	Criteria define success; constraints limit the solution.
C	Inverted operation	Swaps constraint and criterion.
D	Confusion set	Hands-on is not the test; the goal is a design, not an explanation.

5. Answer A Concept 3 · transfer

A teacher wants a science question rather than an engineering problem. Which fits?

Target word: choose the science question — reverse of labeling the engineering task **Decides:** Science seeks to explain the natural world.

A	Correct	Seeks an explanation of a natural phenomenon.
B	Inverted operation	A design problem — engineering.
C	Confusion set	Choosing materials for a design is engineering.
D	Adjacent category	A problem needing a designed solution.

6. Answer D Concept 4

Fifth graders ~~at a school with a large garden~~ want to find whether fertilizer helps bean plants grow taller. One group gives Plant A fertilizer and puts it by the window; Plant B gets no fertilizer and sits in a closet. ~~The group's poster won a ribbon last year.~~ What is the main problem with the plan?

Target word: flaw in the investigation **Decides:** Fertilizer and light both change.

A	Dropped element	Drops the need to hold other variables constant.
B	Absolute language	More trials help, but the main flaw is the second variable.
C	Camouflage	Uses the garden detail; the source of plants is not the flaw.
D	Correct	Light must stay the same; only fertilizer should change.

7. Answer C Concept 4 · transfer

Students test whether ramp height affects how far a toy car rolls. Which should they keep the same?

Target word: identify controlled variables **Decides:** Everything except the variable being tested.

A	Absolute language	Fair tests change one thing at a time.
B	Confusion set	That is the outcome being measured.
C	Correct	These must stay constant so only height changes.
D	Inverted operation	That is the variable they are testing — it must change.

8. Answer B Concept 5

During a pond study ~~on a warm spring morning~~, a first grader writes several notes. ~~The class is using new hand lenses.~~ Which note is an observation rather than an inference?

Target word: pick the observation **Decides:** What is directly seen, heard, or measured.

A	Adjacent category	Adds an explanation — an inference.
B	Correct	Directly observed with the senses.
C	Confusion set	An inference about a cause.
D	Camouflage	Uses the lens detail; an explanation, and a wrong one.

9. Answer A Concept 5 · transfer

A student observes that the sidewalk is wet and the sky is gray. Which statement is an inference based on these observations?

Target word: produce the inference — reverse of spotting the observation **Decides:** An explanation drawn from the observations.

A	Correct	An explanation reasoned from the wet ground and gray sky.
B	Absolute language	Overreaches the evidence.
C	Dropped element	Restates an observation; no reasoning added.
D	Inverted operation	That is an observation.

10. Answer C Concept 6

During a discussion about why ice melts faster on a metal tray than a plastic one, a third grader says, “The metal is colder, so it melts ice faster.” The teacher, ~~who has taught for 15 years, has a bell to signal cleanup.~~ Which response best supports productive talk?

Target word: talk move **Decides:** Press for reasoning and invite others to respond.

A	Camouflage	Uses the bell detail; no talk move.
B	Adjacent category	Praise without probing leaves the idea unexamined.
C	Correct	Presses for evidence and opens student-to-student discussion.
D	Dropped element	Drops the student’s reasoning; ends the discussion.

11. Answer B Concept 7

A fifth-grade class ~~in a new building with modern lab tables~~ will launch small rubber-band rockets made of paper. ~~The teacher has done similar activities for years and no one has been hurt.~~ What is the best safety step?

Target word: safety decision **Decides:** Projectiles call for eye protection and procedures before launch.

A	Camouflage	Past luck is not a safety measure.
B	Correct	Hazards are addressed before the activity.
C	Inverted operation	Safety after an incident is backward.
D	Absolute language	Hands-on work is safe with proper precautions.

12. Answer C Concept 8

Before a unit on seasons, a fourth-grade teacher ~~who loves astronomy~~ asks students to explain why summer is warmer. Many say Earth is closer to the sun in summer. ~~The class has 24 students.~~ What should the teacher do next?

Target word: respond to prior ideas **Decides:** Elicit ideas, then provide experiences that challenge them.

A	Dropped element	Drops the experience that lets students revise their idea.
B	Camouflage	Uses the astronomy interest; unrelated to the misconception.
C	Correct	Making ideas visible and challenging them with evidence supports conceptual change.
D	Absolute language	Misconceptions are durable without direct challenge.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Phenomenon-driven three-dimensional learning	Three-dimensional lessons are anchored in an observable phenomenon students work to explain by using practices, core ideas, and crosscutting concepts together.	Starting with vocabulary and a topic, with the phenomenon as an afterthought.	"How should the teacher launch the unit?"
2	The 5E sequence	Engage, Explore, Explain, Elaborate, Evaluate — students explore before the teacher explains.	Explaining the concept first, then letting students confirm it.	A lesson plan whose phases need ordering.
3	Science vs. engineering practices	Science asks questions and builds explanations about the natural world; engineering defines problems with criteria and constraints and designs and tests solutions.	Calling any hands-on build "science," or skipping criteria and constraints.	A task to design, build, or improve something.
4	Fair tests and controlled variables	A fair test changes one variable at a time and keeps all others the same, so any difference can be traced to that variable.	Changing several things at once.	A student investigation plan with more than one change.
5	Observation vs. inference	An observation comes directly from the senses or instruments; an inference is an explanation drawn from observations.	Recording an inference as an observation.	A student statement that includes "because" or a cause.
6	Productive science talk	Talk moves — revoicing, asking students to add on, and pressing for evidence ("What makes you think that?") — build reasoning and student-to-student discourse.	Evaluating every answer as right or wrong and moving on.	A teacher response to a student's idea during discussion.
7	Safety comes first	Safety equipment, procedures, and hazard review come before and during every hands-on activity, not after an incident.	Treating goggles or rules as optional for "simple" or familiar materials.	A lab involving heat, chemicals, glass, projectiles, or flying pieces.
8	Eliciting and building on prior ideas	Students arrive with everyday ideas; teachers elicit them first, then use experiences that challenge them, so students revise their thinking.	Telling the correct answer and assuming the old idea is gone.	A common misconception surfacing at the start of a unit.

Course scope (public description): "Elementary Science and Engineering Methods" is a dynamic and forward-thinking course designed for educators who aim to integrate science and engineering practices into their elementary classroom. This course focuses on the foundational methods of teaching science and engineering concepts, emphasizing hands-on, experiential learning. Students explore effective strategies for fostering inquiry, creativity, and critical thinking among young learners through science experiences, engineering projects, and the use of technology. The curriculum covers key topics such as the three-dimensional learning approach and scientific literacy and the nature of science. Through intentional instructional decision-making and reflective practice,

educators learn to create an inclusive, stimulating, and safe learning environment that encourages students to explore, question, innovate, and participate in productive discourse in the fields of science and engineering.